PATRICK BAILLIE ISLES

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SYSTEMS & INFRASTRUCTURE ENGINEER | LINUX KERNEL & SILICON DIAGNOSTICS | ZERO-ALLOCATION RUST

Systems-focused engineer combining advanced analytical research in formal logic, epistemology, and primary-source analysis with low-level systems programming. Proven record diagnosing silicon-to-kernel failure modes, authoring upstream kernel driver tooling, and engineering high-frequency, zero-allocation telemetry daemons across 10 live Prometheus endpoints (:9106–:9115). Specializes in raw IOCTLs, kernel buddy allocation & compaction, x86_64 assembly, and deterministic root-cause forensics.

CORE TECHNICAL COMPETENCIES

Languages:	**Rust** (2021 Edition, Multi-threading, Zero-Copy, Inline ASM, FFI, POSIX Signals), **Python 3** (Typing, AST), **Bash/POSIX Shell**, C (Foundational), SQL
Linux Systems & Kernel:	Buddy Allocator & THP Direct Compaction, VFS Page Cache Writeback, Raw NVMe Admin IOCTL, SysV IPC Reaping, Block T10-PI (DIF/DIX), POSIX RT Scheduling, x86_64 `RDTSC` Hardware Latency, ACPI/PROCHOT, `systemd`
Networking & Protocols:	Linux SocketCAN (`AF_CAN`), SAE J1939-21 / CAN 2.0B, UART/Serial (115200 8N1), TCP/IP, Kerberos V5, OpenLDAP (RFC 2307)
Security & Enterprise IAM:	Kerberos KDC (AES-256), SSSD (LDB Dynamic Caching, HBAC), PAM/NSS, SELinux MLS Policies, Kickstart Automation
Observability & Telemetry:	**10 Prometheus Telemetry Daemons (:9106–:9115)**, Grafana, HWLAT Micro-sampling, DSP (True RMS, Ripple), Formal Systems Modeling

SYSTEMS ENGINEERING & OPEN-SOURCE PORTFOLIO

Upstream Linux Kernel & Systems Runtime (rustix • signal-hook • socketcan-rs) ***Rust, Raw Syscalls, Linux SocketCAN, POSIX Signals, C-FFI, sigaltstack***

- Authored upstream `socketcan-rs` PRs **#108** (kernel `can-isotp`) and **#107** (J1939). Resolved 5-year Tokio bug (**tokio#3520**) in `signal-hook-registry` adding `SA_ONSTACK` across POSIX targets, eliminating fatal Go runtime aborts.
- Patched `bytecodealliance/rustix` (**#1730**), eliminating startup panics in `page_size()` under Linux capabilities (`setcap`) on kernels < 6.4 via zero-allocation dynamic `mmap`/`mprotect` alignment probing.

Buddy Allocator Compaction & VFS Page Cache (buddy-compact-sentinel • dirty-throttle-sentinel) ***Rust, Linux Buddy Allocator, THP Compaction, VFS***

- **Buddy Allocator & THP Compaction Sentinel**: Interrogated `/proc/buddyinfo` and `/proc/vmstat` to compute mathematical external fragmentation indices (R_{frag}) for Order 9 (2MB HugePages); alerted on Sev-0 synchronous Direct Compaction stalls (`compact_stall`) that freeze application threads for 500ms+ (`:9115`).
- **VFS Writeback Choke Auditor**: Interrogated `/proc/sys/vm/` and `/proc/vmstat` queues to compute distance-to-throttle headroom before synchronous `TASK_UNINTERRUPTIBLE` (`D` state) write freezes; exposed via Prometheus (`:9110`).

Ring -2 SMM Stolen Time & Silicon PROCHOT Sentinels (smi-storm-sentinel • silicon-throttle-rs) ***Rust, x86 RDTSC, Ring -2 SMM, PROCHOT***

- **Ring -2 SMM Firmware Auditor**: Executed raw x86_64 `_rdtsc()` assembly in a lock-free micro-sampling loop to detect invisible hardware latency gaps where BIOS/UEFI firmware abducts the CPU into SMRAM without kernel awareness (Prometheus `:9114`).
- **Silicon Clock-Stretching & PROCHOT**: Interrogated CPU frequency scaling nodes and ACPI cooling devices to expose the "Phantom Slowdown"—where thermal downclocking slashes compute speed by up to 8x while masking as 100% busy in standard APM tools (`:9112`).

Block Layer T10-PI & POSIX Real-Time Schedulers (blk-integrity-sentinel • rt-sched-sentinel) ***Rust, T10 DIF/DIX, Storage Integrity, POSIX RT***

- **Block Data Integrity (DIF/DIX)**: Interrogated `/sys/block/*/integrity/` auditing 8-byte T10-PI sector protection tags (Guard CRC, Ref LBA) to expose storage volumes running in Type 0 Unprotected Mode vulnerable to silent data corruption (SDC) and torn writes (`:9113`).
- **Real-Time Sched & Priority Inversion**: Interrogated `SCHED_FIFO` tasks for Mars Pathfinder-style priority inversion futex deadlocks and nanosecond runqueue waiting latency (`run_delay`) via `/proc/[pid]/schedstat` (Prometheus `:9111`).

Silicon NVMe 2.0 IOCTL & System V IPC Reapers (nvme-sentinel-rs • kmem-drain-rs) ***Rust, Raw IOCTL, SysV IPC, Slab Leaks***

- **NVMe Hardware Telemetry**: Interrogated NVMe controllers via raw `NVME_IOCTL_ADMIN_CMD`, decoding 512-byte packed SMART logs, endurance wear, composite temperature, and PCIe ASPM link degradation (`:9108`).
- **IPC Memory Reaper**: Interrogated `/proc/sysvipc/shm` to detect orphaned dead-PID shared memory segments and unreclaimable slab leaks (`SUnreclaim`); implemented automated purging via `shmctl(IPC_RMID)` (`:9109`).

EDUCATION & FORMAL CREDENTIALS

Electronic Engineering Technology & Mathematics Track • Portland Community College **Honors: Presidents' List (4.0 Term GPA), Multiple Deans' List**

Curriculum & Mastery: Higher-level Calculus Series (Calculus I: MTH 251, Calculus II: MTH 252), Technical Writing (WR 227), Honors Public Speaking (COMM 111H: A), Electronic Circuit Analysis. Advanced through higher-level calculus with documented academic honors.

Bachelor of Arts (B.A.) in Philosophy • Minor in History (Awarded: March 2020) • Portland State University **International Study: Heidelberg University (Germany)**

International Study: Ruprecht-Karls-Universität Heidelberg • German Language & Continental Philosophy

Advanced Analytical & Research Specialization:

- **Formal Logic & Deduction (PHL 324U):** Propositional/predicate calculus, formal validity, and truth functional semantic models.
- **Historical Thesis & Research Methodology (HST 492 / HST 495):** 400-level research seminars focused on primary-source critical analysis, historiography, and empirical argumentation.
- **Epistemology & Philosophy of Language (PHL 424 / PHL 433):** Formal models of truth verification, linguistic syntax, semantics, and structural demarcation.
- **German Technical Translation Series (GER 101–301):** Multilingual proficiency in technical German and international academic correspondence.